

# Sample document

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## 1 LAlignAnd

$$\begin{array}{ll} \&= & \begin{array}{l} a = b \\ c = d \end{array} \quad \text{ok} \\ \\ =\& & \begin{array}{l} a = b \\ c = d \end{array} \quad \text{ng} \\ \\ =\{\}\& & \begin{array}{l} a = b \\ c = d \end{array} \quad \text{ok} \end{array}$$

$$\begin{array}{ccccccc} x = y & x \leq y & x \leq y & x < y & x > y & x \geq y & x \approx y \\ x \neq y & x \geq y & x \geq y & x > y & x < y & x \leq y & x \asymp y \end{array}$$

verbatim does not care about this rule: `a =\& b`

## 2 LAlignEnd

The following ends with a line break.

$$f(x) = ax^2 + bx + c \tag{1}$$

$$g(x) = dx^2 + ex + f \tag{2}$$

(3)

The following does not end with a line break.

$$f(x) = ax^2 + bx + c \tag{4}$$

$$g(x) = dx^2 + ex + f \tag{5}$$

Here is the next line after the align environment.

### 3 LAlignSingleLine

|                                                                                                                      |                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| — Long line before display (same result) —                                                                           |                                                                                                                  |
| Lorem ipsum.<br><br>$f(x) = ax^2 + bx + c$<br><br>This is an <span style="color: blue;">equation</span> environment. | Lorem ipsum.<br><br>$f(x) = ax^2 + bx + c$<br><br>This is an <span style="color: red;">align</span> environment. |

|                                                                                                              |                                                                                                          |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| — Short line before display (different result) —                                                             |                                                                                                          |
| Lrm:<br><br>$f(x) = ax^2 + bx + c$<br><br>This is an <span style="color: blue;">equation</span> environment. | Lrm:<br><br>$f(x) = ax^2 + bx + c$<br><br>This is an <span style="color: red;">align</span> environment. |

Single-line alignat environment is also detected.

$$f(x) = ax^2 + bx + c$$

Multi-line alignat environment is not detected.

$$f(x) = ax^2 + bx + c$$

$$g(x) = dx^2 + ex + f$$

### 4 LLArticle

A  $n$ -dim. An  $n$ -dim. a  $n$ -dim. an  $n$ -dim. A  $n$  dim. An  $n$  dim. a  $n$  dim. an  $n$  dim.

#### 4.1 Single Letters

OK: an  $a$ -dim, a  $b$ -dim. a  $u$ -dim.

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• NG: a <math>a \rightarrow</math> OK: an <math>a</math></li> <li>• NG: a <math>e \rightarrow</math> OK: an <math>e</math></li> <li>• NG: a <math>f \rightarrow</math> OK: an <math>f</math></li> <li>• NG: a <math>h \rightarrow</math> OK: an <math>h</math></li> <li>• NG: a <math>i \rightarrow</math> OK: an <math>i</math></li> <li>• NG: a <math>l \rightarrow</math> OK: an <math>l</math></li> </ul> | <ul style="list-style-type: none"> <li>• NG: a <math>m \rightarrow</math> OK: an <math>m</math></li> <li>• NG: a <math>n \rightarrow</math> OK: an <math>n</math></li> <li>• NG: a <math>o \rightarrow</math> OK: an <math>o</math></li> <li>• NG: a <math>r \rightarrow</math> OK: an <math>r</math></li> <li>• NG: a <math>s \rightarrow</math> OK: an <math>s</math></li> <li>• NG: a <math>x \rightarrow</math> OK: an <math>x</math></li> </ul> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

NG: an  $u \rightarrow$  OK: a  $u$

## 4.2 Abbreviations

- NG: a EM  $\rightarrow$  OK: an EM (Expectation–Maximization)
- NG: a EVD  $\rightarrow$  OK: an EVD (Eigenvalue Decomposition)
- NG: a FFT  $\rightarrow$  OK: an FFT (Fast Fourier Transform)
- NG: a NP-hard  $\rightarrow$  OK: an NP-hard (Non-deterministic Polynomial-time hard)
- NG: a LSTM  $\rightarrow$  OK: an LSTM (Long Short-Term Memory)
- NG: a LTI  $\rightarrow$  OK: an LTI (Linear Time-Invariant)
- NG: a MLE  $\rightarrow$  OK: an MLE (Maximum Likelihood Estimation)
- NG: a MSE  $\rightarrow$  OK: an MSE (Mean Squared Error)
- NG: a ODE  $\rightarrow$  OK: an ODE (Ordinary Differential Equation)
- NG: a RNN  $\rightarrow$  OK: an RNN (Recurrent Neural Network)
- NG: a RKHS  $\rightarrow$  OK: an RKHS (Reproducing Kernel Hilbert Space)
- NG: a SDE  $\rightarrow$  OK: an SDE (Stochastic Differential Equation)
- NG: a SVD  $\rightarrow$  OK: an SVD (Singular Value Decomposition)
- NG: a SVM  $\rightarrow$  OK: an SVM (Support Vector Machine)
- NG: a XOR  $\rightarrow$  OK: an XOR (Exclusive OR)

## 4.3 LaTeX Commands

OK: An  $\mathbb{R}$ -valued.

- |                                                                            |                                                                                      |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| • NG: ‘a $\mathbb{R}$ -valued’ $\rightarrow$ OK: ‘an $\mathbb{R}$ -valued’ | • NG: ‘a $\ell^2$ norm’ $\rightarrow$ OK: ‘an $\ell^2$ norm’                         |
| • NG: ‘a $L^1$ norm’ $\rightarrow$ OK: ‘an $L^1$ norm’                     | • NG: ‘a $\ell^\infty$ norm’ $\rightarrow$ OK: ‘an $\ell^\infty$ norm’               |
| • NG: ‘a $L^2$ norm’ $\rightarrow$ OK: ‘an $L^2$ norm’                     | • NG: ‘a $\mathcal{L}^1$ norm’ $\rightarrow$ OK: ‘an $\mathcal{L}^1$ norm’           |
| • NG: ‘a $L^\infty$ norm’ $\rightarrow$ OK: ‘an $L^\infty$ norm’           | • NG: ‘a $\mathcal{L}^2$ norm’ $\rightarrow$ OK: ‘an $\mathcal{L}^2$ norm’           |
| • NG: ‘a $\ell^1$ norm’ $\rightarrow$ OK: ‘an $\ell^1$ norm’               | • NG: ‘a $\mathcal{L}^\infty$ norm’ $\rightarrow$ OK: ‘an $\mathcal{L}^\infty$ norm’ |

## 5 LLBig

This is a sample text. This is a sample text. This is a sample text.

Both bigcup  $\bigcup_{x \in B} O_x$  and cup  $\cup_{x \in B} O_x$  do not spoil the line spacing.

This is a sample text. This is a sample text. This is a sample text.

$$\begin{array}{ccccccccc}
X_1 \cap X_2 & X_1 \cup X_2 & X_1 \odot X_2 & X_1 \oplus X_2 & X_1 \otimes X_2 & & & & \\
X_1 \sqcup X_2 & X_1 \uplus X_2 & X_1 \vee X_2 & X_1 \wedge X_2 & \text{ok} & & & & \\
X_1 \oplus_{\mathbb{R}} X_2 & X_1 \oplus_R X_2 & \text{ok} & & & & & & \\
\bigcap_{i=1}^{\infty} X_i & \bigcup_{i=1}^{\infty} X_i & \bigodot_{i=1}^{\infty} X_i & \bigoplus_{i=1}^{\infty} X_i & \bigotimes_{i=1}^{\infty} X_i & \bigsqcup_{i=1}^{\infty} X_i & \biguplus_{i=1}^{\infty} X_i & \bigvee_{i=1}^{\infty} X_i & \bigwedge_{i=1}^{\infty} X_i \quad \text{ok} \\
\cap_{i=1}^{\infty} X_i & \cup_{i=1}^{\infty} X_i & \odot_{i=1}^{\infty} X_i & \oplus_{i=1}^{\infty} X_i & \otimes_{i=1}^{\infty} X_i & & & & \\
\sqcup_{i=1}^{\infty} X_i & \uplus_{i=1}^{\infty} X_i & \vee_{i=1}^{\infty} X_i & \wedge_{i=1}^{\infty} X_i & \text{ng} & & & & 
\end{array}$$

## 6 LLBracketCurly

$$\begin{array}{lll}
\backslash\max(a,b) & \max(a,b) & \text{ok} \\
\backslash\max\{a,b\} & \max a,b & \text{ng} \\
\backslash\max \{a,b\} & \max a,b & \text{ok?}
\end{array}$$

We cannot fully determine whether the use of curly brackets is wrong or not. It is not detected if some spaces are inserted between the command name and the curly brackets.  $\min(a,b)$  and  $\min a,b$  are also checked.

## 7 LLBracketMissing

$$\begin{array}{lll}
x^{\{23\}} & x^{23} & \text{ok} \\
x^{\{2\}3} & x^23 & \text{ok} \\
x^{23} & x^23 & \text{ng}
\end{array}$$

$x_23$ ,  $x^ab$ , and  $x_ab$  are also checked. Cases like  $x^ab$ ,  $x^2$ , and  $e^i\pi$  are not detected. Disabled by default. (It contains false positives, and may be too strict for some users.) Following examples are ignored.

```

escaped underscore: \_123
https://sample_url.com
\includegraphics{sample_link.png}
\includegraphics{sample_link.pdf}
\label{sample_label}
\cref{sample_label}
\eqref{sample_label}
\cite{sample_label}
\Cref{sample_Label}
\bibliographystyle{my_style}
\addbibresource{ref_file.bib}
% x^23 comment

```

## 8 LLBracketRound

|                       |              |    |
|-----------------------|--------------|----|
| <code>\sqrt{a}</code> | $\sqrt{a}$   | ok |
| <code>\sqrt(a)</code> | $\sqrt{(a)}$ | ng |

$a^{(1)}$  and  $a_{(1)}$  are also checked.  
 Following examples are ignored:  
`Test\_ (1)` `Test\_ (1)`  
`Test\^2` `Test^2`  
`sample_(1).png` `sample_(1).pdf`



Figure 1: sample\_(1).png

`\label{eq:f(x_(k+1)) <= m(x_(k+1)) <= m(x_k) = f(x_k)}`

$$f(x_{k+1}) \leq m(x_{k+1}) \leq m(x_k) = f(x_k) \quad (6)$$

The label in Eq. (6) is ignored.

## 9 LLColonEqq

|                        |                     |    |
|------------------------|---------------------|----|
| <code>\coloneqq</code> | $x := y$            | ok |
| <code>\Coloneqq</code> | $x \mathrel{::}= y$ | ok |
| <code>:=</code>        | $x := y$            | ng |
| <code>::=</code>       | $x \mathrel{::}= y$ | ng |

The difference is quite subtle, but the vertical position of the colon is different. This difference become clearer with some other math fonts.

## 10 LLColonForMapping

|                        |                                         |     |
|------------------------|-----------------------------------------|-----|
| <code>A:B</code>       | $A : B$                                 | ok  |
| <code>A\colon B</code> | $A : B$                                 | ok? |
| <code>f:</code>        | $f : \mathbb{R} \rightarrow \mathbb{R}$ | ng  |
| <code>f\colon</code>   | $f : \mathbb{R} \rightarrow \mathbb{R}$ | ok  |

— We detect all of : in the following —

Here are examples of colons we detect.

- $f : X \rightarrow Y$
- $g : X \mapsto Y$
- $h : \mathbb{R}^{n^2+2n+1} \rightarrow \mathbb{R}$

and

$$f : (X \text{ at new line in tex file}) \rightarrow (Y \text{ at new line in tex file}). \quad (7)$$

— We do NOT detect any of : in the following —

Here are examples of ‘:’ we do not detect.

- $f:X \rightarrow Y$ , the correct use of `\colon`.
- $A : B : C = 1 : 2 : 3$ , the colon for ratio.
- $A : B = 1 : 2$  and  $X \rightarrow Y$ , separated by dollar sign.
- $g : (\text{some very very very very very long long long long words}) \rightarrow \mathbb{R}$ , the false negative.

The following contains colon:  $f : X \rightarrow Y$

The following contains colon:  $f : X \rightarrow Y$

The following contains colon:

$$f : X \rightarrow Y \quad (8)$$

$$\lim_{k \rightarrow \infty} x_k = L \quad (9)$$

## 11 LLCref

**Theorem 1.** *This is a sample theorem.*

Use Thm. 1 with cref instead of Thm. 1 with ref to avoid mistakes. Disabled by default.

---

**Algorithm 1** Sample Algorithm

---

- 1: First line
  - 2: Second line
- 

In the pseudocode, `cleveref` can fail to appropriately reference lines. Thus, we don’t detect `\ref{line:1}` like this: Line 1. We detect other formats `\ref{myline:2}` like this: Line 2. You can change these behaviors via settings.

## 12 LLDots

|                       |                 |    |
|-----------------------|-----------------|----|
| <code>\dots</code>    | $x, \dots, y$   | ok |
| <code>\ldots</code>   | $x, \dots, y$   | ok |
| <code>, \dots,</code> | $x, \dots, y$   | ng |
| <code>\dots</code>    | $x + \dots + y$ | ok |
| <code>\cdots</code>   | $x + \dots + y$ | ok |
| <code>, \dots,</code> | $x + \dots + y$ | ng |
| <code>\dots</code>    | 0.333...        | ok |
| <code>\dotso</code>   | 0.333...        | ok |
| <code>...</code>      | 0.333...        | ng |

Unicode: `.....`, dots: `...`, textellipsis: `...`, dotso: `...`

## 13 LLDoubleQuotes

Use “XXX” instead of “XXX” or ”XXX”. You can use them for `H\”older` and `\verb.`  
“Checking LLText...”

“double quotes in verbatim”

```
def test():
    S = "double quotes in listing"
```

## 14 LLENDash

|                            |     |
|----------------------------|-----|
| <code>hyphen (-)</code>    | A-B |
| <code>en-dash (--)</code>  | A–B |
| <code>em-dash (---)</code> | A—B |

Examples:

- Erdos-Renyi (random graph, Erdős–Rényi)
- Einstein-Podolsky-Rosen (quantum physics, Einstein–Podolsky–Rosen)
- Fruchterman-Reingold (graph drawing, Fruchterman–Reingold)
- Gauss-Legendre (numerical integration, Gauss–Legendre)
- Gibbs-Helmholtz (thermodynamics, Gibbs–Helmholtz)
- Karush-Kuhn-Tucker (optimization, Karush–Kuhn–Tucker)

Exceptions: Non-Negative, Well-Known, etc.

## 15 LLEqnarray

We should not use eqnarray. It has some spacing issues.

$$\begin{array}{rcl} x & = & y \\ a & = & b \end{array}$$

## 16 LLErrCompOps

$$\begin{array}{cccccccccccc} x \leq y & x \geq y & x = y & \text{ok} & & & & & & & & \\ x <= y & x <= y & x < y & x < y & x \leq y & x \leq y & x \leq y & x \leq y & \text{ng} & & & \\ x >= y & x >= y & x > y & x > y & x \geq y & x \geq y & x \geq y & x \geq y & \text{ng} & & & \end{array}$$

## 17 LLFootnote

### 17.1 Bad Examples

Space before <sup>1</sup>

Tab before <sup>2</sup>

Line break before <sup>3</sup>

Line break before with comment <sup>4</sup>

Space before percent sign <sup>5</sup>

### 17.2 Good Examples

Correct attach.<sup>6</sup>

Phrase attach<sup>7</sup> ok.

Line break suppressed<sup>8</sup> Line break suppressed<sup>9</sup>

Inserted comment<sup>10</sup>

Emphasis *text*<sup>11</sup>.

---

<sup>1</sup>bad footnote

<sup>2</sup>bad footnote

<sup>3</sup>bad footnote

<sup>4</sup>bad footnote

<sup>5</sup>bad footnote

<sup>6</sup>good footnote

<sup>7</sup>good footnote

<sup>8</sup>good footnote

<sup>9</sup>good footnote

<sup>10</sup>good footnote

<sup>11</sup>good footnote

## 18 LLHeading

### 18.0.1 Incorrect Hierarchy

### 18.1 First Subsection

#### 18.1.1 Second Subsubsection

## 19 LLLlGg

\ll     $n \ll m$     ok  
<<     $n < m$     ng

I like human <<< cat <<<<<<< dog.

## 20 LLMathPunctuation

$$f(x) = (x + a)(x + b) \tag{10}$$

$$= x^2 + (a + b)x + ab, \tag{11}$$

The last comma should be a period.

$$e^{i\pi} + 1 = 0.$$

which is called Euler's identity.

## 21 LLNonASCII

The following line contains non-ASCII characters.

! " # \$ % & ' ( ) \* + - /

日本語の文章は、upLaTeXでフツウに書けます。

(You can write Japanese sentences as usual with upLaTeX.)

## 22 LLNonstandard

### 22.1 Therefore and Because

Not Standard?:  $\therefore a = b, \because x > 0$

Standard: Therefore,  $a = b$ . Because  $x > 0$ , ...

### 22.2 Iff

Not Standard?: This statement is true iff the condition holds.

Standard: This statement is true if and only if the condition holds.

OK: \iff is a valid LaTeX command and should not be detected.  $a = b \iff c = d$ .

## 22.3 DotSeq

Not Standard?:  $a \doteq b, x \doteq y$

Standard:  $a \approx b, x \approx y$

## 22.4 Combination Notation

Not Standard?:  ${}_nC_k, {}_nC_k$

Standard:  $\binom{n}{k}$

## 23 LLOperator

|                                        |                           |     |
|----------------------------------------|---------------------------|-----|
| <code>2\operatorname{rank}A</code>     | $2\operatorname{rank}A$   | ok  |
| <code>2\operatorname{rank} A</code>    | $2\operatorname{rank}A$   | ok  |
| <code>2\,\operatorname{rank}\,A</code> | $2\operatorname{rank}A$   | ok? |
| <code>2\operatorname{rank} A</code>    | $2\operatorname{rank}A$   | ng  |
| <code>n_{\operatorname{rank}}</code>   | $n_{\operatorname{rank}}$ | ok  |

- $\dim A$  vs  $\dim A$
- $\operatorname{rank} A$  vs  $\operatorname{rank} A$
- $\operatorname{tr} A$  vs  $\operatorname{tr} A$
- $\det A$  vs  $\det A$
- $\arg z$  vs  $\arg z$
- $\operatorname{Re} z$  vs  $\operatorname{Re} z$
- $\operatorname{Im} z$  vs  $\operatorname{Im} z$

## 24 LLPeriod

The period character (.) can be used in abbreviations, not as a sentence-ending period. In the former case, it should be followed by a backslash to prevent LaTeX from treating it as a sentence-ending period, which can affect spacing.

|                          |                              |    |
|--------------------------|------------------------------|----|
| <code>e.g., Test.</code> | $e.g., \operatorname{Test}.$ | ok |
| <code>e.g.\ Test.</code> | $e.g. \operatorname{Test}.$  | ok |
| <code>e.g. Test.</code>  | $e.g. \operatorname{Test}.$  | ng |

Without a backslash: e.g. blah i.e. blah a.k.a. blah i.i.d. blah w.r.t. blah w.l.o.g. blah resp. blah etc.

With a backslash: e.g. blah i.e. blah a.k.a. blah i.i.d. blah w.r.t. blah w.l.o.g. blah resp. blah etc.

## 25 LLRefEq

To refer to the equation, use (7) with eqref instead of (7) with ref.  
Eq. 7 and Prob. 7 are allowed.

## 26 LLSharp

Some people mistake `\sharp` to be a symbol for the number of elements in a set, but it is not.  
It is a musical symbol. The correct symbol for the number of elements in a set is `\#`.

|                     |                             |    |
|---------------------|-----------------------------|----|
| <code>\#</code>     | $\#\{x \mid x \geq 0\}$     | ok |
| <code>\sharp</code> | $\sharp\{x \mid x \geq 0\}$ | ng |

- matching examples:  $\#A$ ,  $\#\{1, 2, 3\}$ ,  $\#X_0$
- non-matching examples:  $\sharp$ ,  $f^\sharp X$ ,  $\alpha_\sharp X$ ,  $C^\sharp$

If you really want to use  $\sharp$ , you can disable this rule.

## 27 LLSI

|                                 |      |    |
|---------------------------------|------|----|
| <code>\SI{1}{\kilo\byte}</code> | 1 kB | ok |
| 1 kB                            | 1 kB | ng |
| 1kB                             | 1kB  | ng |

10KB, 3.5 MiB, 500GB are detected. 123 noNumWord GB will not be detected.  
Some command named as EB. This is not ExaByte. This 1EB is one ExaByte.

## 28 LLSortedCites

Unsorted citations [2, 1]: ng?

Automatically sortable like this: [1, 2].

This rule is skipped for documents that use the `subfiles` document class.

## 29 LLSpaceEnglish

Missing spaces around inline math should be detected.

Improper spacing:  $xyz$  and  $xyz$ . Proper spacing:  $x\ y\ z$  and  $x\ y\ z$ .

The following are excluded by rule exceptions:  $nth$ ,  $mth$ ,  $\backslash n$ ,  $\backslash m$ ,  $H_2O$ ,  $C_nH_{2n+2}$ .

No detection in the following cases:  $1x$ ,  $1x$ ,  $x1$ ,  $x1$ ,  $x$ ,  $x$ , etc.

## 30 LLSpaceJapanese

The following are Japanese examples, which are excluded by rule exceptions.

日本語の文章で  $x = 1$  もしくは  $y = 2$  と数式を書くと、スペースが欠如します。

日本語の文章で  $x = 1$  もしくは  $y = 2$  と数式を書くと、スペースが生まれます。

尤も、フォーマルな文章では非推奨な場合もあり、その為デフォルトでは無効です。  
 テスト:これは大丈夫。テスト(これは大丈夫)。1文字だけ間にある場合として $x$ と $y$ のテスト。

## 31 LLT

|                                                       |          |     |
|-------------------------------------------------------|----------|-----|
| $\text{\textasciitilde}\backslash\text{top}$          | $X^\top$ | ok  |
| $\text{\textasciitilde}\{\text{\textsf{mathsf{T}}}\}$ | $X^\top$ | ok  |
| $\text{\textasciitilde}T$                             | $X^T$    | ng  |
| $\text{\textasciitilde}\{T\}$                         | $X^T$    | ok? |

Disabled by default. The following cases are not detected as errors.

$$\sum_{i=1}^T x_i, \quad \sum_0^T y_i, \quad \sum_{\alpha}^T z_i, \quad \sum_{i=1}^T y_i,$$

$$\prod_{i=1}^T x_i, \quad \prod_0^T y_i, \quad \prod_{\beta}^T z_i, \quad \prod_{i=1}^T y_i,$$

$$\int_0^T f(t)dt, \quad \int_0^T f(t)dt, \quad \int_{\gamma}^T f(t)dt$$

## 32 LLTextLint

Only for Japanese texts.

ご静聴ありがとうございました。 御静聴ありがとうございました。 まず第1に、背景を説明します。 まず第一に、背景を説明します。 このモデルをシュミレーションします。 この命題は帰納法で証明できます。 1番最初に確認します。 一番最初に確認します。 一番最後に結論を述べます。 この実装はまだ未完成です。 開催日時はまだ未定です。 この問題はまだ未解決です。 これは第1回目です。 今日は第2日目です。 この関数の2回導関数を考えます。

Sample text: Hamaguchi et. al. provide supplemental materials on Youtube, Github and VScode Marketplace, which contains Latex source code. This sentence contains errors we will detect by LLTextLint.

## 33 LLThousands

|                 |        |    |
|-----------------|--------|----|
| $\$12, 34\$$    | 12, 34 | ok |
| 1,000           | 1,000  | ok |
| $\$1\{,\}000\$$ | 1,000  | ok |
| $\$1,000\$$     | 1,000  | ng |

## 34 LLTitle

### 34.1 This Is a Correct Title

#### 34.1.1 this is a wrong title

The quick brown fox jumps over the lazy dog

SubParagraph: Test With Ref 1

### 34.2 IGNORE IF ALL UPPERCASE

### 34.3 Math Contains version $x$

## 35 LLUnRef

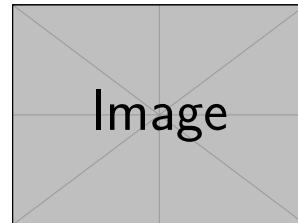
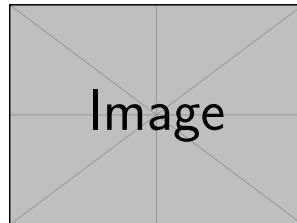
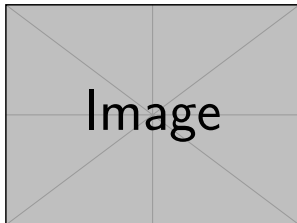


Figure 2: Referenced Figure   Figure 3: Unreferenced Figure   Figure 4: Referenced Figure 2

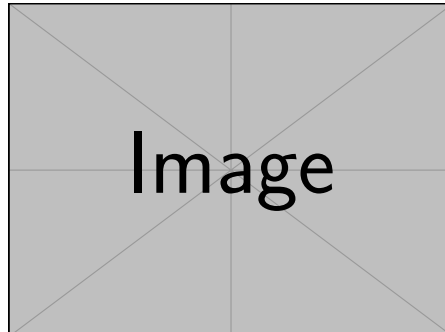


Figure 5: Unreferenced Wide Figure

| A | B |
|---|---|
| 1 | 2 |

Table 1: Referenced Wide Table

Figure 2, fig. 4, and 1 are referenced in the text, while the other figure/table labels are not.

|   |   |
|---|---|
| C | D |
| 3 | 4 |

Table 2: Unreferenced Wide Table

## 36 LLURL

The following URLs contain unnecessary query strings (should be detected):

- `https://example.com/page?utm_source=newsletter&utm_medium=email`
- `https://example.com/page?sessionid=123456`

The following URLs contain only allowed query strings (should not be detected):

- `https://example.com/search?q=latex`
- `https://example.com/list?page=2`
- `https://example.com/view?lang=ja`

Examples of URLs without query strings:

- `https://example.com/clean-page`

## 37 LLUserDefined

You can define your own rule.

|                              |          |    |
|------------------------------|----------|----|
| <code>f^\mathrm{a}(x)</code> | $f^a(x)$ | ok |
| <code>f^a(x)</code>          | $f^a(x)$ | ng |

|                           |               |    |
|---------------------------|---------------|----|
| <code>f \infConv g</code> | $f \square g$ | ok |
| <code>f \Box g</code>     | $f \square g$ | ng |

## References

- [1] A. Article. *Journal*, 2000.
- [2] B. *Book*. Publisher, 2000.

## Appendix A Check LLText

Wrong LaTeX math environment:  
 $\begin{align}$   
 $\end{equation}$

## Appendix B LLSetBar

Detecting inappropriate use of the vertical bar  $|$  is very difficult. We are currently trying to detect the following, although not implemented yet.

|                    |                    |    |
|--------------------|--------------------|----|
| $\lvert -1 \rvert$ | $\lvert -1 \rvert$ | ok |
| $\abs{-1}$         | $\lvert -1 \rvert$ | ok |
| $\vert -1 \vert$   | $\lvert -1 \rvert$ | ng |
| $  -1  $           | $\lvert -1 \rvert$ | ng |

|                    |                    |    |
|--------------------|--------------------|----|
| $\lVert -x \rVert$ | $\lVert -x \rVert$ | ok |
| $\norm{-x}$        | $\lVert -x \rVert$ | ok |
| $\Vert -x \Vert$   | $\lVert -x \rVert$ | ng |
| $   -x   $         | $\lVert -x \rVert$ | ng |

|                       |                                           |     |
|-----------------------|-------------------------------------------|-----|
| $\relmiddle $ (macro) | $\left\{ a \mid a > \frac{1}{2} \right\}$ | ok  |
| $\mid$                | $\{ a \mid a > \frac{1}{2} \}$            | ok? |
| $ $                   | $\{ a   a > \frac{1}{2} \}$               | ng  |

|                       |              |     |
|-----------------------|--------------|-----|
| $\divides$ (MnSymbol) | $+2 \mid +4$ | ok  |
| $\mid$                | $+2 \mid +4$ | ok? |
| $\mathrel{\mid}$      | $+2 \mid +4$ | ok? |
| $\vert$               | $+2 \mid +4$ | ng  |
| $ $                   | $+2 \mid +4$ | ng  |

|                                          |                                          |     |
|------------------------------------------|------------------------------------------|-----|
| $f(y x)$                                 | $f(y x)$                                 | ok? |
| $f(y \mid x)$                            | $f(y \mid x)$                            | ok? |
| $f(\backslash,y\mid x\backslash,)$       | $f(y \mid x)$                            | ok? |
| $\left. \frac{d}{dt} \right _{t=0} f(t)$ | $\left. \frac{d}{dt} \right _{t=0} f(t)$ | ok? |